

Stanley College of Engineering & Technology for Women (A)**B.E (CSE/CME/AI&DS/IT) IV Semester Main Examination -July-2023****Operating Systems****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. List out different services of operating systems.
- b. What is a thread? Name the benefits of thread.
- c. Classify the necessary conditions for Deadlock.
- d. What is virtual memory?
- e. Describe page fault.
- f. Mention the key evaluation points of RAID System.

[2M CO1 BTL1]

[2M CO2 BTL1]

[2M CO3 BTL2]

[2M CO4 BTL1]

[2M CO4 BTL2]

[2M CO5 BTL1]

Part-B**4X12=48M**

2. a) Define OS and explain functions of OS.
- b) What is system call? Identify various types of system calls in detail.
3. a) From the below given data. Draw Gantt charts illustrating the execution of the processes using FCFS and SJF scheduling algorithms and calculate the average turnaround time and waiting time for them.

[6M CO1 BTL2]

[6M CO1 BTL3]

[6M CO2 BTL2]

Processes	Arrival Time	Burst Time
P1	0	4
P2	1	2
P3	2	2
P4	3	5
P5	4	6

- b) Define process. Draw and explain various Process States in detail.
4. a) Illustrate about critical-section problem and Peterson's solution in concurrency.
- b) How to apply the monitors to solve the dining philosopher's problem with example?
5. a) Differentiate external fragmentation with internal fragmentation.
- b) Distinguish about Logical and Physical Addressing.
6. Consider the following page reference string:
 7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1
 Assuming the demand paging with three frames, determine how many page faults would occur for the following replacements algorithms?
 a. LRU b. FIFO c. Optimal
7. a) Summarize about file attributes, file operations, and file types.
- b) Discuss any two of disk scheduling algorithms with reference blocks.

[6M CO2 BTL3]

[6M CO3 BTL2]

[6M CO3 BTL3]

[6M CO4 BTL4]

[6M CO4 BTL4]

[12M CO4 BTL5]

[6M CO5 BTL2]

[6M CO5 BTL6]

Hall Ticket No. –

Code: R123744

Stanley College of Engineering & Technology for Women (A)

B.E (IT) IV Semester (Main) Examinations-July-2023

Fundamentals of Digital Image Processing

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. What do you meant by Gray level? [2M, CO1, BTL1]
b. Analyze the need of color image processing. [2M, CO1, BTL4]
c. Write the equations for discrete Fourier transform for one-dimensional and two-dimensional function. [2M, CO2, BTL1]
d. What is image negative? [2M, CO3, BTL2]
e. Compare image enhancement and restoration techniques. [2M, CO4, BTL4]
f. Distinguish lossless and lossy compression. [2M, CO5, BTL4]

Part-B

4X12=48M

2. a) Explain by drawing diagram fundamental steps in image processing. [6M, CO1, BTL1]
b) Illustrate the concept of neighbors of a pixels and adjacency. [6M, CO1, BTL2]
3. a) Determine the Kernel coefficient of slant transform for $N=4$. [7M, CO2, BTL3]
b) What are the properties and advantages of Haar Transform? [5M, CO2, BTL2]
4. a) Analyze the expressions in the histogram equalization technique for image enhancement. [6M, CO3, BTL4]
b) Explain the following order static filters
i) Median filter ii) Min and max filter. [6M, CO3, BTL2]
5. a) Explain model of image degradation/restoration process with a block diagram. [6M, CO4, BTL2]
b) How Wiener filtering is helpful to reduce the mean square error? [6M, CO4, BTL4]
6. a) Explain how to detect discontinuities in image segmentation. [5M, CO5, BTL4]
b) Define image compression. Explain about the redundancies in a digital image. [7M, CO5, BTL3]
7. a) List out different color models in color image processing and discuss about pseudo color image processing. [6M, CO1, BTL2]
b) With necessary plots explain the following intensity transformation techniques:
i) Gray level slicing ii) Logarithmic transformation [6M, CO3, BTL3]

Stanley College of Engineering & Technology for Women (A)

B.E (IT) IV Semester (Main) Examinations-July-2023

Theory of Automata

Time: 03 Hours

Max. Marks-60

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Part-A

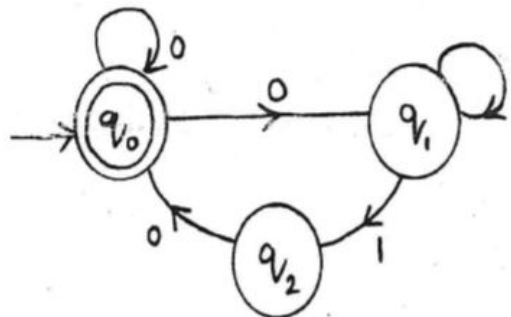
6X2=12M

1. a. How to check acceptance of string by finite automata? [2M CO2 BTL1]
- b. What is a Regular expression in the theory of Automata? [2M CO3 BTL1]
- c. For the Grammar $\{S \rightarrow AS/a, A \rightarrow SbA/SS/ba\}$ construct Left most derivation for the string aabbbaaa. [2M CO3 BTL6]
- d. Write about the model of Push Down Automata. [2M CO4 BTL1]
- e. What is Turing machine halting problem? [2M CO5 BTL1]
- f. Why is chomsky hierarchy important? Draw a neat sketch. [2M CO5 BTL1]

Part-B

4X12=48M

2. a) Show with an example equivalence between NFA with and without ϵ - transitions. [6M CO1 BTL4]
- b) Define NFA mathematically. Explain its significance and function. Convert the given Finite automaton into its Deterministic equivalence. Explain method used. Taking suitable example and prove both accept the same string.



[6M CO1 BTL4]

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3. a) Construct a DFA for the Regular Language consisting of any number of a's and b's.
[5M CO2 BTL5]
- b) What is a regular language? Convert the given regular expression to regular language.
i) $(1+\epsilon)(00^*1)0^*$
ii) $(0^*1^*)000(0+1)^*$
iii) $(00+10)^*1^*(10+00)^*$
[7M CO2 BTL3]
4. a) Define Context Free Grammar. State and Explain the closure properties of CFG.
[5M CO2 BTL3]
- b) Define Ambiguous Grammar? Check whether the grammar $S \rightarrow Aab$, $A \rightarrow Bc/cd$,
 $C \rightarrow cd$, $B \rightarrow c/d$ Is Ambiguous or not?
[7M CO4 BTL3]
5. a) Define Push Down Automata. Explain the basic structure of PDA with a neat
Graphical representation.
[6M CO4 BTL3]
- b) Why the Languages accepted by Turing machine are called "recursively enumerable languages". Explain at least two Closure properties of recursively enumerable languages.
[6M CO4 BTL3]
6. a) Explain in detail about PCP with examples.
[7M CO4 BTL1]
- b) How can the Halting problem of Turing machine be Handled? Explain. [5M CO4 BTL3]
7. Construct a context free grammar for a given language and construct the PDA for the
same $L = \{ 0^n 1^n / n \geq 1 \}$
[12M CO4 BTL3]

Stanley College of Engineering & Technology for Women (A)**B.E (IT) IV Semester (Main) Examinations-July-2023****Computer Organization and Microprocessor****Time: 03 Hours****Max. Marks-60**

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Part-A**6X2=12M**

1. a. What is Fixed-point representation? [2M CO1BTL1]
 b. Give examples of Address Sequence. [2M CO2 BTL1]
 c. Write the advantage of RISC over CISC? [2M CO3 BTL3]
 d. Define an I/O processor. [2M CO4 BTL1]
 e. List out functions of DMA. [2M CO5 BTL1]
 f. What is ADC? Give an example. [2M CO5 BTL1]

Part-B**4X12=48M**

2. a) Discuss the bus and memory transfer. [4M CO1 BTL4]
 b) What is register transfer language? Explain the basic symbols used in register transfer. [8M CO1 BTL3]
3. a) Explain the instruction cycle with the help of a flow chart. [6M CO1 BTL2]
 b) How to design the control unit? Explain. [6M CO2 BTL6]
4. a) What is virtual memory? With a neat block diagram explain the virtual memory address translation. [6M CO3 BTL5]
 b) Briefly explain the floating-point arithmetic operations with examples. [6M CO3 BTL3]
5. a) Why does DMA have priority over the CPU when both request a memory transfer? [4M CO5 BTL2]
 b) What are addressing modes? Explain different types of addressing. [4M CO5 BTL2]
6. a) Discuss the architecture of the 8051 microcontrollers. [4M CO3 BTL2]
 b) Explain the features of multi core processors. [4M CO3 BTL2]
7. a) Write a short note on GPUs. [6M CO4 BTL2]
 b) Explain the Programmable Interrupt Controller 8259. [6M CO5 BTL3]

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Hall Ticket No. -

Code: R123754

Stanley College of Engineering & Technology for Women (A)

B.E (IT) IV Semester (Main) Examinations-July-2023

Techniques on Signals and Systems

Time: 03 Hours

Max. Marks-60

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Part-A

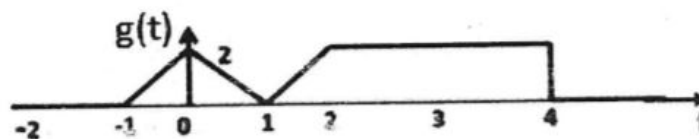
6X2=12M

1. a. Sketch the following signal $x(t) = u(t+1) - 2u(t) + u(t-1)$ [2M CO1 BTL2]
b. State the Parseval's power theorem of Fourier series. [2M CO2 BTL1]
c. Find the Fourier transform of signal $x(t) = 5\sin^2 3t$ [2M CO2 BTL3]
d. Determine the DTFT of the sequence $x(n) = 0.5^n u(n)$ [2M CO2 BTL3]
e. Define Region of Convergence. Also, state any two properties of the Region of convergence with respect to Z Transform. [2M CO4 BTL1]
f. Determine whether the following system is stable $h(t) = te^{-t}u(t)$ [2M CO3 BTL4]

Part-B

4X12=48M

2. a) Categorize the given signal as Energy or Power Signal $x(t)3[u(t) - u(t-5)]$ [6M CO1 BTL4]
b) For the signal shown in Fig.1, sketch the following: [6M CO1 BTL3]
(i) $g(2t-2)$ (ii) even part of $g(t)$ (iii) odd part of $g(t)$



3. a) Consider a rectified sine wave signal $x(t)$ defined by $x(t) = |A \sin \Pi t|$ [6M CO1 BTL3]
i) Sketch $x(t)$ and find its fundamental period.
ii) Find the complex exponential Fourier series of $x(t)$
b) State the Dirichlet conditions of Fourier series. [6M CO2 BTL1]
4. a) State and prove Frequency differentiation and Frequency shift properties of Fourier transforms. [7M CO2 BTL2]
b) Interpret the Fourier Transform of the signal using appropriate properties. [5M CO2 BTL5]
 $x(t) = e^{-2t} \sin(\Pi t) u(t)$

5. a) Find the inverse Laplace transform of the following function

[6M CO3BTL3]

$$F(s) = \frac{s+1}{(s+1)^2 + 4}$$

b) A Laplace transform is characterized by the differential equation

[6M CO3 BTL6]

$$\frac{d^2}{dt^2}y(t) + \frac{5dy(t)}{dt} + 6y(t) = x(t)$$

Solve for $y(t)$ for $t \geq 0$ when $x(t) = u(t)$ $y(0^-) = 2$ and $\frac{dy(0^-)}{dt} = -12$

6. a) List any three properties of Laplace Transform.

[5M CO3 BTL1]

b) State and Prove the following properties of DFT

[7M CO2 BTL2]

i) Multiplication in time domain ii) Complex conjugate

7. a) Simplify the given function and calculate the inverse Z-transform

[6M CO4BTL4]

$$X(z) = \frac{z}{3z^2 - 4z + 1} \quad \text{for the ROCs}$$

i) $|z| > 1, |z| < \frac{1}{3}$

ii) $\frac{1}{3} < |z| < 1$

b) A system is described by the difference equation LTI

[6M CO3BTL3]

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$$

- i) Find the transfer function of the system.
- ii) Find the impulse response of the system $h(n)$
- iii) Find the step response of the system $s(n)$